



Manaranjan Studys

Chapter 7 — System of Particles & Rotational Motion

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What You'll Learn in Chapter 7

This chapter shifts focus from single-particle motion to systems of many particles and rigid bodies — introducing the powerful concepts of center of mass, torque, angular momentum, and rotational dynamics.

01

Center of Mass

Position, velocity & acceleration of COM for particle systems

02

Linear Momentum

Conservation laws for systems of particles

03

Vector Product

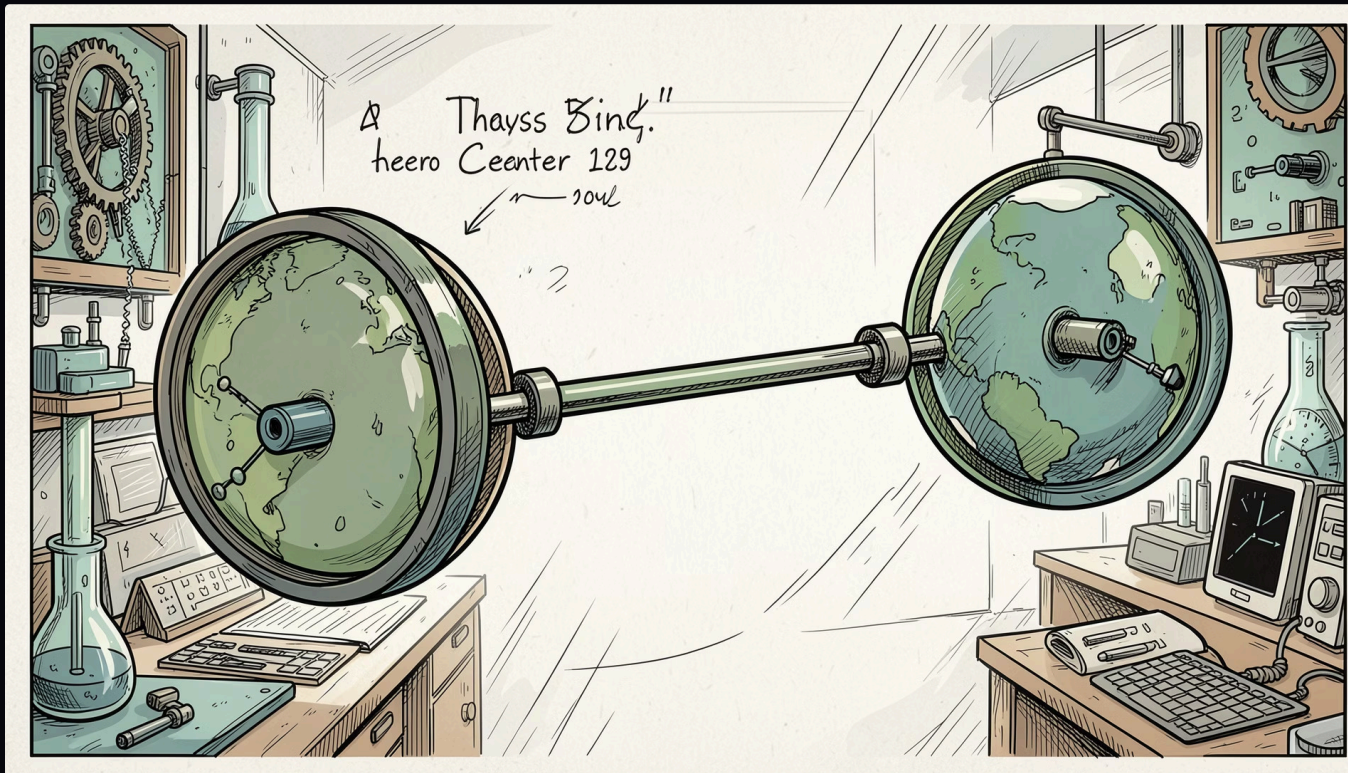
Cross product, torque & angular momentum

04

Rotational Motion

Moment of inertia, theorems & rolling motion

Center of Mass (COM)



Definition

The center of mass is the point where the entire mass of a system can be assumed to be concentrated for analyzing translational motion.

Two-Particle System

$$X_{cm} = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}$$

N-Particle System

$$\vec{R}_{cm} = \frac{\sum m_i \vec{r}_i}{\sum m_i}$$

- ① For a uniform rigid body, COM lies at its geometric center.

Motion of Center of Mass

The COM of a system moves as if all external forces act on a single particle of mass M located at the COM.

Velocity of COM

$$\vec{V}_{cm} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2 + \dots}{M}$$

Acceleration of COM

$$\vec{a}_{cm} = \frac{\vec{F}_{ext}}{M}$$

Conservation of Momentum

If $\vec{F}_{ext} = 0$, then $\vec{P}_{total} = \text{constant}$ and $\vec{V}_{cm} = \text{constant}$.

Vector (Cross) Product & Torque

Cross Product

$$\vec{A} \times \vec{B} = AB \sin \theta \hat{n}$$

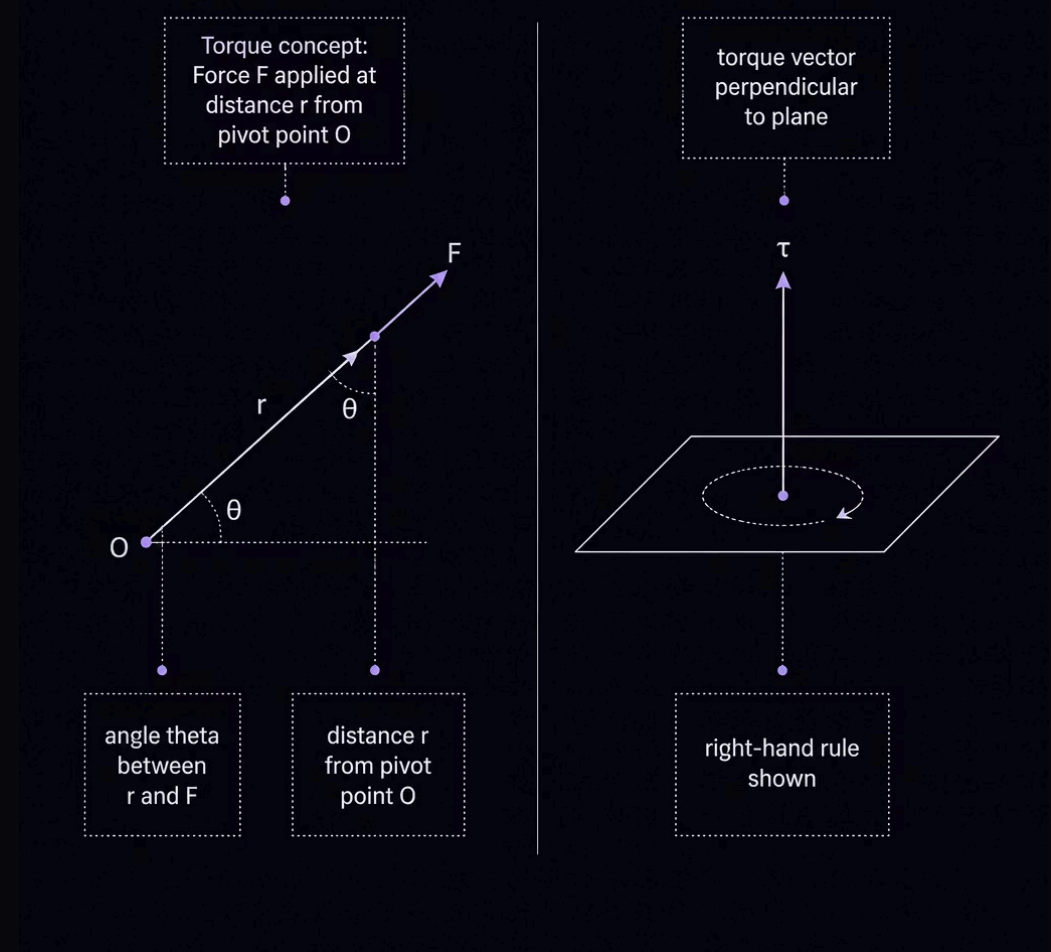
Result is a vector perpendicular to the plane of $\vec{A}$ and $\vec{B}$ (right-hand rule).

Torque

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\tau = rF \sin \theta$$

SI Unit: $\text{N} \cdot \text{m}$ • Dimensions: $[ML^2T^{-2}]$



Angular Momentum

Definition

$$\vec{L} = \vec{r} \times \vec{p} = m(\vec{r} \times \vec{v})$$

Magnitude

$$L = mvr \sin \theta$$

Relation with Torque

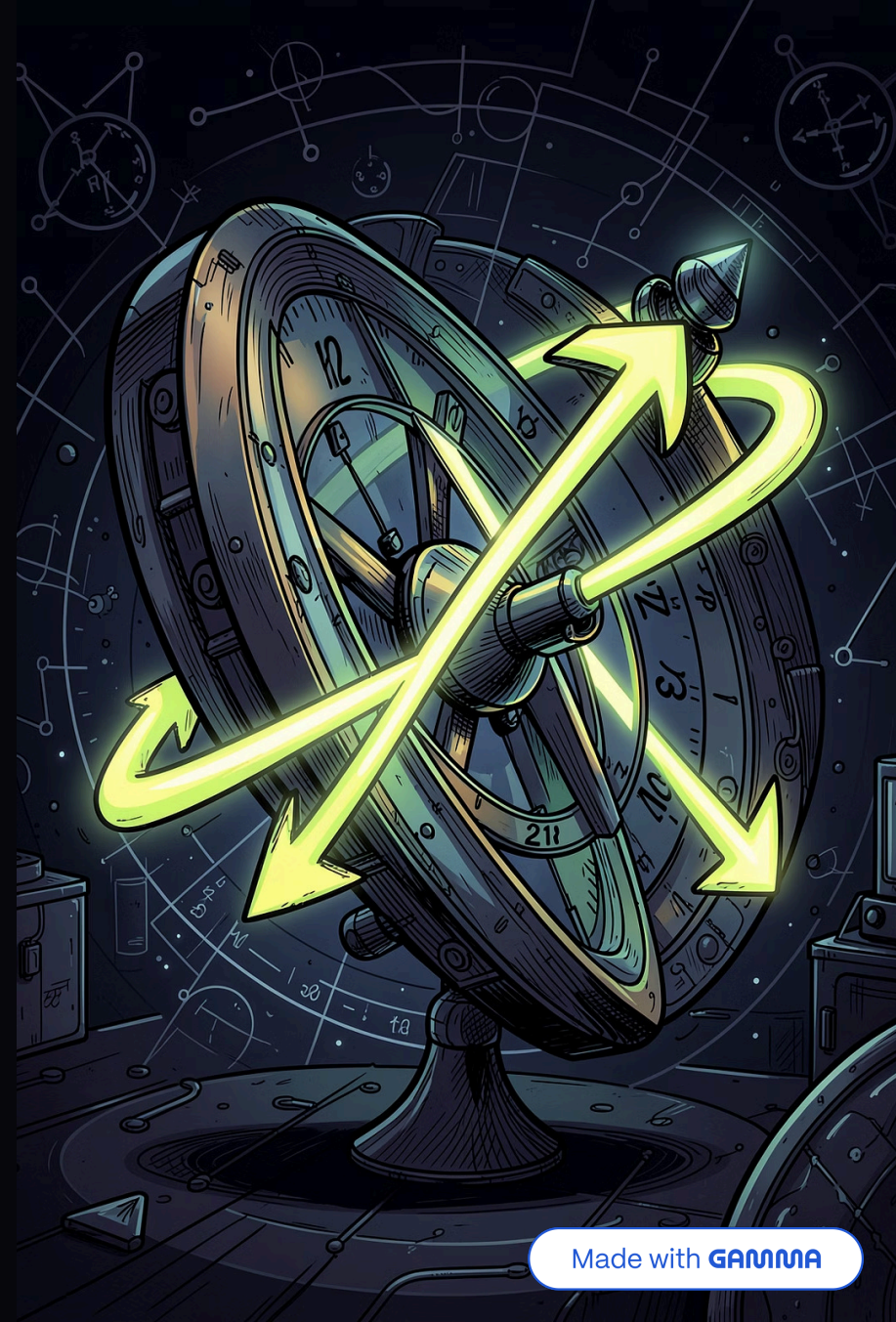
$$\vec{\tau} = \frac{d\vec{L}}{dt}$$

Conservation Law

If $\vec{\tau}_{ext} = 0$, then $\vec{L} = \text{constant}$.

SI Unit

$\text{kg} \cdot \text{m}^2/\text{s}$ • Dimensions:
 $[ML^2T^{-1}]$



Moment of Inertia & Radius of Gyration

Moment of Inertia

$$I = \sum m_i r_i^2$$

Rotational analog of mass. Depends on mass distribution and axis of rotation.

Radius of Gyration (K)

$$I = MK^2 \implies K = \sqrt{\frac{I}{M}}$$

Distance from axis where entire mass can be assumed concentrated.

Body	I (about axis)
Thin ring (center)	MR^2
Solid disc (center)	$\frac{1}{2}MR^2$
Solid sphere (diameter)	$\frac{2}{5}MR^2$
Hollow sphere	$\frac{2}{3}MR^2$
Rod (center, $\perp$)	$ML^2/12$

Theorem of Perpendicular & Parallel Axes

⊥ Axes Theorem

For a **plane lamina**:

$$I_z = I_x + I_y$$

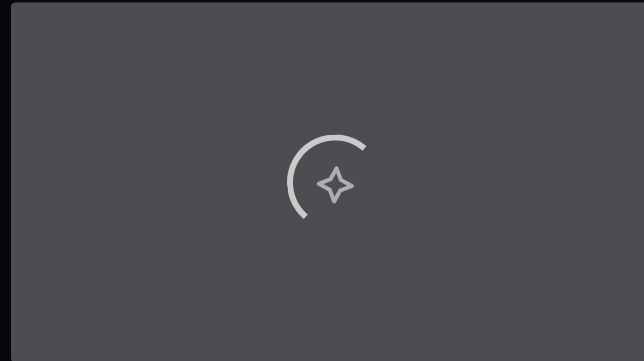
Where z-axis is perpendicular to the plane containing x and y axes.

Parallel Axes Theorem

For **any rigid body**:

$$I = I_{cm} + Md^2$$

Where I_{cm} is about COM and d is distance between axes.



Rotational Dynamics & Rolling Motion

Newton's 2nd Law (Rotational)

$$\tau = I\alpha$$

Analogous to $F = ma$

Rotational Kinetic Energy

$$K.E. = \frac{1}{2}I\omega^2$$

Energy due to rotation

Rolling Without Slipping

$$v_{cm} = R\omega$$

Condition for pure rolling

Total KE (Rolling)

$$K = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$$

Translation + Rotation

❏ **Key Insight:** For rolling bodies, friction does *no work* in pure rolling — it only enables rotation.

Chapter 7 — Formula Sheet at a Glance

1

Center of Mass

$$\vec{R}_{cm} = \frac{\sum m_i \vec{r}_i}{M}$$

2

Torque

$$\vec{\tau} = \vec{r} \times \vec{F}$$

3

Angular Momentum

$$\vec{L} = \vec{r} \times \vec{p}$$

4

Moment of Inertia

$$I = \sum m_i r_i^2$$

5

Parallel Axes

$$I = I_{cm} + Md^2$$

6

Rolling Condition

$$v = R\omega$$